NOISEMAP · FEASIBILITY ASSESSMENT

NoiseMap feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-02	Amish Chadha	First issue (kit 1.8 pipeline)

NoiseMap feasibility assessment

The pole clamp node (option A) is feasible as a paper design: the physics and arithmetic hold (NSM-STM-001, all 24 claims within known science, 3 of them closed by calculation), it is modelled at detail level D3 with 89 parts and no overlaps, and its parts cost is USD 72.00 against a USD 100 target. What cannot be settled on paper is measurement quality: accuracy, range at the quiet end and frequency response are rated at risk and need a class 1 reference meter at TRL 4, which is on hold. Method: STANDARDS section 23.

Table 1. Feasibility by aspect (option A unless stated).

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	Levels computed in the head, a FieldNode core for power and radio, an arm to put the microphone 4.0 m up and 0.45 m off the pole. Dimensions and interfaces are modelled and 62 fit checks pass	High	Mass margin is 0.05 kg (R10, now PR-010): any added part needs a mass check	NSM-STM-001 S10 to S12, NSM-CAL-001 H, cad/fidelity.json
Measurement quality	Microphone data support 27.9 to 119 dB SPL with the IM72D128 fitted by default (103 to 108 dBA with the ICS-43434) and A and C weighting after equalizing; expanded uncertainty 2.9 dB on assumed terms against 2 dB	Medium	Membrane, windscreen and head diffraction are unknown; facade bias up to 2.4 dB before a site correction; these need a reference meter	NSM-CAL-001 A to C, NSM-PRS-001 PR-003 to PR-005
Materials	Aluminium sections, stainless bands and fixings, printed ASA head, polycarbonate box, open-cell foam	High	Foam and membrane ageing; galvanic couple is small by area ratio	NSM-STM-001 S15, S16
Manufacturing and assembly	V-blocks sawn from 16 mm bar, saddle bent from 2.5 mm sheet, head printed in ASA; a making sketch for each made part and a picture for each step	High	Needs sawing, bending, printing and fine soldering; no certified trade; bought tube flange dimensions to confirm when parts are bought	NSM-BLD-001, NSM-DDR-003
Power and controls	12.2 mW load, 5.81 Wh/day harvest at 1.5 sun hours, 53 days without sun; no feedback loop	High	Cold-weather charging relies on FieldNode's lockout	NSM-STM-001 S5, S6, S17, NSM-CAL-001 D and E
Radio	22 byte record, 23.7 s/day at SF9, interval rule at slow data rates	High	Needs gateway coverage; fair use is a community rule	NSM-STM-001 S7, S8, S18
Supply of parts	The ICS-43434 is listed end of life by its maker; the IM72D128 is the alternative on the same adapter board	Medium	Single sources for the modules; prices indicative (September 2026 estimates)	bom/bom.csv, NSM-DDR-001 D3
Regulatory	No certification claimed; GDPR, radio and battery items to assess	Medium	See NSM-REG-001	NSM-REG-001
Concept options B and C	Both are feasible as designs (STM S20 to S24). B is cheap and simple but measures at a facade; C needs mains inside and a lighting operator	B Medium, C Low	C fails the no-electrical-connection constraint of the pitch	NSM-OPT-001

Ratings: High, Medium, Low.